Coal Mining, Drinking Water Quality, and the Value of Remediation

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The views expressed are the authors' and do not necessarily reflect the views or policies of the U.S. Environmental Protection Agency.

Background

Coal mining in the US has long been known to harm water quality.

- Documented in many detailed, narrow-in-scope studies such as: Corbett, 1977; Powell, 1988; Lambert et al., 2004; Griffith et al., 2012; Brooks et al., 2019

Coal mining is also the subject of long-standing, prominent environmental regulation.

Clean Water Act (CWA Effluent Guidelines est. 1975)

- Coal preparation plants, water circuits, and associated areas
- Acid, alkaline drainage from active mines
- "Post-mining areas"

Surface Mining Control and Reclamation Act (SMCRA) of 1977

- Established the Office of Surface Mining Reclamation and Enforcement (DOI)
- Enforces reclamation bond requirements and funds mine remedition

Research Questions

What is the impact of coal mining on (drinking) water quality?

What is the value of remediation projects, such as those required by the SMCRA of 1977?

To answer these questions, we combine three elements:

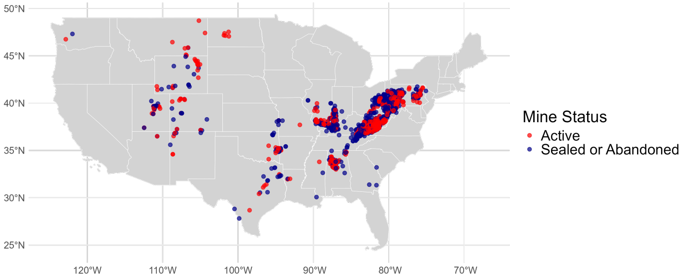
- Comprehensive data on coal-mining operations nationally
- Nationally-representative, researcher-compiled water quality sampling data
- Information on mine remediation projects (in progress)

Data and General Approach (1/2)

Link water quality samples to mining exposure via proximity of intake locations to coal mines.

Coal Mine Information and Operations Data

- Mine Safety and Health Administration (MSHA)
- Annual production 2000-2019, mine closure, technology
- More than 95% of total U.S. production represented in data



Data and General Approach (2/2)

Link water quality samples to mining exposure via proximity of intake locations to coal mines.

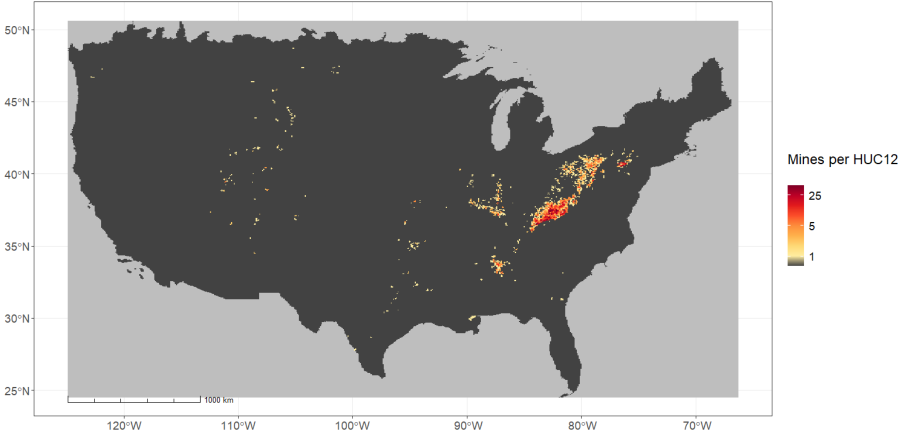
Drinking Water Quality Database

- Analyte samples collected to satisfy Safe Drinking Water Act compliance
- Millions of samples for a wide variety of analytes 2000-2019
- Samples are tied to Public Water System (PWS) intake locations
- Other Inorganic Chemicals (IOCs): Antimony, Arsenic, Barium, Beryllium, Cadmium, Chromium, Copper, Cyanide, Fluoride, Lead, Mercury, Selenium, Thallium

Analyte Group	Min	Median	Mean	P95	Max	Unique PWS	Observations
Disinfectant Byproducts	0.00	18.00	36.04	95.30	1346.00	46,120	3,184,109
Lead and Copper	0.00	1.40	78.74	400.00	4780.00	55,373	4,881,621
Nitrate/Nitrite	0.00	100.00	1755.11	7680.00	600000.00	143,772	7,088,987
Other IOC	0.00	0.00	97.47	740.00	7750.00	67,608	7,683,808

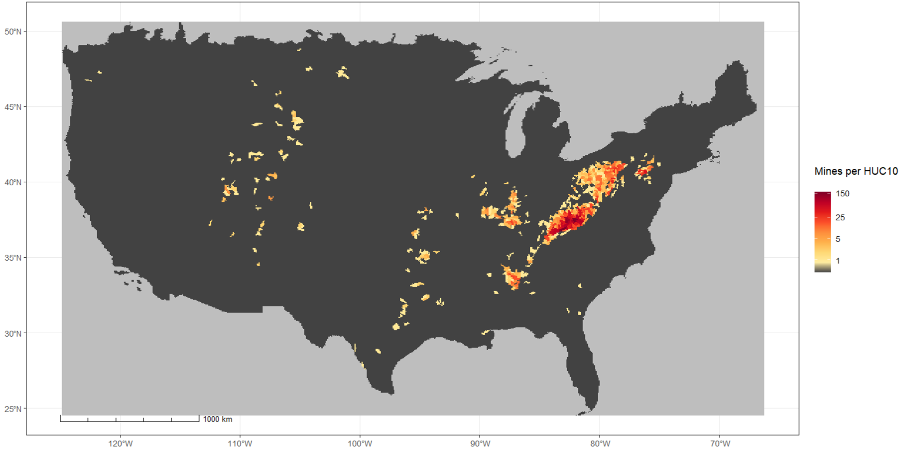
Key Data Linkage

Mining Activity $\rightarrow$ HUC12s $\leftarrow$ PWS intakes



Key Data Linkage

Mining Activity → HUC10s ← PWS intakes



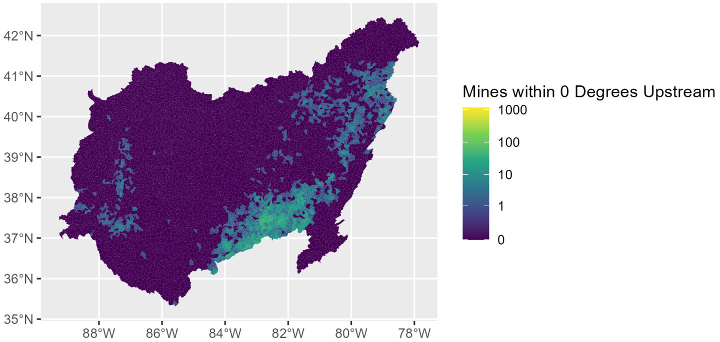
Empirical Strategy

Estimate effect of coal production on downstream water quality

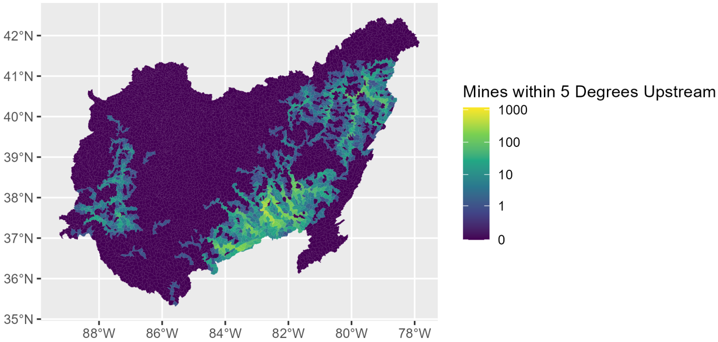
- Outcome variable: percent of samples above EPA's MCL for each analyte
  - Similar to the Keiser et al. (2023) working paper (effect of EPA grants)
- Treatment: For a given PWS-year, the coal production variable is defined as *cumulative production since 2000 in intake HUCs (and upstream HUCs)*
  - Exploits continuous variation in an always-treated group; as compared to a never-treated group
- Other details:
  - PWSID-by-analyte fixed effects: local characteristics of water system, sampling techniques
  - Intake-HUC02-by-year fixed effects: comparisons within hydrologic region

Note: MCL = maximum contaminant level, EPA's enforceable drinking water standard

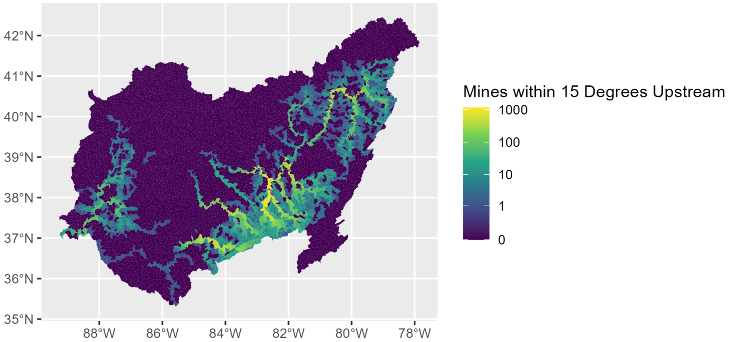
Upstream Variation (0 links)



Upstream Variation (5 links)



Upstream Variation (15 links)



Equation for Causal Analysis

$$P_{sahy} = \beta D_{sy} + \gamma_{hy} + X'_{say}\theta + \mu_{sa} + \epsilon_{say} \tag{1}$$

- Indices: system s, analyte a, hydrologic region of intakes h, year y

- P_{say}: Percent of samples above MCL (EPA's enforceable standard)
- D_{sy}: Cumulative coal production impacting system s intakes since 2000

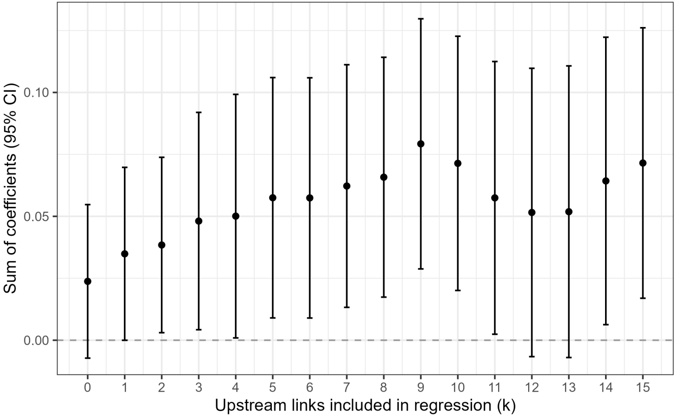
- γ_{hy}: HUC02-Year FE
- X_{say}: Controls (share of samples by month)
- μ_{sa}: System-by-analyte FEs
- ϵ: Error term, clustered by system

Effect of Production on the Percent of Samples Above MCL

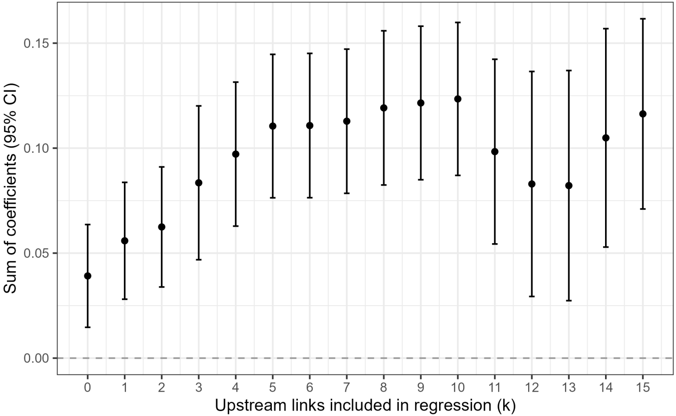
	All	DBPs	Lead and Copper	Nitrate-Nitrites	Other IOCs
	(1)	(2)	(3)	(4)	(5)
Panel A: Only production in same HUC12					
coal_st_cumsum_huc12	0.0238	-0.0492	0.1019*	0.0375**	0.0391***
	(0.0158)	(0.1208)	(0.0544)	(0.0177)	(0.0125)
PWSID x Analyte FE	Yes	Yes	Yes	Yes	Yes
HUC02 x Year FEs	Yes	Yes	Yes	Yes	Yes
Month ctrls.	Yes	Yes	Yes	Yes	Yes
Observations	7,372,788	682,693	529,556	2,632,782	3,527,757
Panel B: Only production in same HUC10					
coal_st_cumsum_huc10	0.0092	-0.0104	0.0266*	0.0140***	0.0248***
	(0.0059)	(0.0397)	(0.0159)	(0.0049)	(0.0045)
PWSID x Analyte FE	Yes	Yes	Yes	Yes	Yes
HUC02 x Year FEs	Yes	Yes	Yes	Yes	Yes
Month ctrls.	Yes	Yes	Yes	Yes	Yes
Observations	7,372,788	682,693	529,556	2,632,782	3,527,757

Each coefficient reports the effect of 10 Million Short Tons of production on the percent of analyte samples above the MCL

Effect of Including Upstream Production (All Samples)



Effect of Including Upstream Production (IOCs)



Concluding thoughts

The “big picture” impact of coal mining on downstream water quality remains unclear, despite being a classic example of both a prominent water pollution externality and a focus of major environmental regulations.

- We provide evidence that coal mining continues to negatively impact downstream water quality across the US
- We find that coal mining elevates concentrations of several IOCs beyond EPA-determined standards for drinking water in nearby public water systems
- We hope to extend this work to better understand sources of heterogeneous impacts, and the effects of public investments in coal mine remediation projects.

Thank you

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